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COLD BOOT ATTACKS, UNENCRYPTED RAM EXTRACTION



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Did you know that even if your system is whole disk encrypted, your data can still be extracted using something called a cold boot attack? Read on.

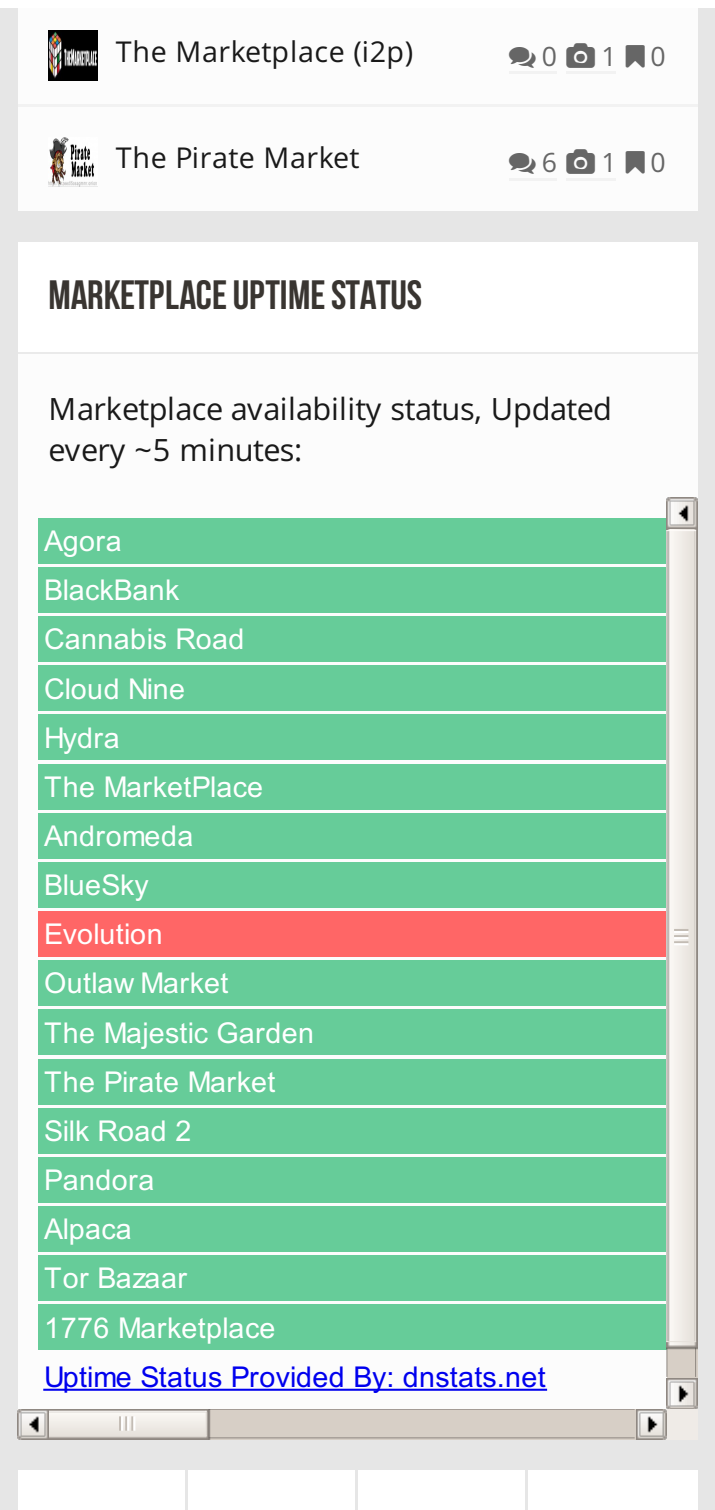
The first thing we need to talk about is RAM. RAM stands for random access memory. All you need to know about RAM is that RAM is the place in a

computer where the operating system, application programs, and data in current use are kept so that they can be quickly reached by the computer's processor. RAM is much faster to read from and write to than the other kinds of storage in a computer, the hard disk, floppy disk, and CD-ROM. However, the data in RAM stays there only as long as your computer is running. When you turn the computer off, RAM loses its data.

When you turn your computer on again, your operating system and other files are once again loaded into RAM, usually from your hard disk. RAM can be compared to a person's short-term memory and the hard disk to the long-term memory. The short-term memory focuses on work at hand, but can only keep so many facts in view at one time. If short-term memory fills up, your brain sometimes is able to refresh it from facts stored in long-term memory. A computer also works this way. If RAM fills up, the processor needs to continually go to the hard disk to overlay old data in RAM with new, slowing down the computer's operation. Unlike the hard disk which can become completely full of data, RAM never runs out of memory.

Data can be extracted from the RAM using various tools. When you have a text document open and you are working on it, you are working from the RAM. Meaning that if you are working on a sensitive document, that document is temporarily stored in the RAM and is vulnerable to being extracted while the computer is on. When RAM is being stored, it is being stored **without** any form of encryption, making it very easy to steal and a huge security risk.

Shutting down a computer through its normal shutdown cycle usually goes through a process of clearing the RAM. However, if the computer loses



power abruptly like in a power outage, the computer does not go through its normal shut down cycle and some information remains on the RAM chips for a few seconds up to a few minutes. This is one of the ways cold boot attacks can work.

I also want to quickly introduce a type of RAM to you which will help you understand the rest of this article better. Below is a research paper and they used a type of ram called DRAM. DRAM stands for **dynamic random access memory**. DRAM is the most common kind of random access memory (RAM) for personal computers and workstations. DRAM is dynamic in that, unlike static RAM (SRAM), it needs to have its storage cells refreshed or given a new electronic charge every few milliseconds. DRAM is designed to lose its memory quickly after losing power. Then there are subsections of DRAM called DDR. This is a way of making the memory more quickly available, but it is not really important to fully understand. Wikipedia can give you all you need to know about DDR. In this article we are focusing on just the concept of DDR, DDR2 and DDR3.

These are newer versions of DRAM that keep getting better, and I believe we are currently up to DDR4. But most computers circulating around today have DDR2 and DDR3 in them unless they are older computers, this includes laptops. DRAM is known as a type of volatile memory, it is computer memory that requires power to maintain the stored information. It retains its contents while powered, but when power is interrupted, stored data is quickly lost. But how quickly is it lost?

In 2008, a group of researchers wanted to see the practicality of extracting unencrypted data from the RAM in your computer. They argued that DRAMs

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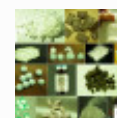
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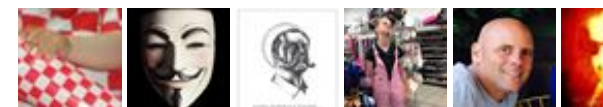
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used in most modern computers retain their contents for seconds to minutes after power is lost, even at operating temperatures and even if removed from a motherboard. And by using an analysis tool they were able to search for key files (such as PGP keys) held in the RAM that could be used to decrypt encrypted volumes (drives) on your computer. They successfully were able to decrypt volumes using BitLocker, FileVault, dm-crypt, and TrueCrypt. Below is the abstract of their research.

Quote

“ “ Lest We Remember: Cold Boot Attacks on Encryption Keys

Abstract Contrary to popular assumption, DRAMs used in most modern computers retain their contents for seconds to minutes after power is lost, even at operating temperatures and even if removed from a motherboard. Although DRAMs become less reliable when they are not refreshed, they are not immediately erased, and their contents persist sufficiently for malicious (or forensic) acquisition of usable full-system memory images. We show that this phenomenon limits the ability of an operating system to protect

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ability of an operating system to protect cryptographic key material from an attacker with physical access. We use cold reboots to mount attacks on popular disk encryption systems — BitLocker, FileVault, dm-crypt, and TrueCrypt — using no special devices or materials. We experimentally characterize the extent and predictability of memory remanence and report that remanence times can be increased dramatically with simple techniques. We offer new algorithms for finding cryptographic keys in memory images and for correcting errors caused by bit decay. Though we discuss several strategies for partially mitigating these risks, we know of no simple remedy that would eliminate them.

<https://citp.princeton.edu/research/memory/>
[Abstract] <http://citpsite.s3-website-us-east-1.amazonaws.com/oldsite-htdocs/pub/coldboot.pdf> [Full Text]

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Here is an FLV video you can download to watch exactly how they did it.

<https://anonfiles.com/file/97b5737dba6b96871fd862b8a587b8f0>

This was very troubling to most people, and had many people freaking out when the research paper was released back in 2008 because even tough encryption tools like TrueCrypt could be rendered useless with an attack like this. Upon further analysis of the paper, I wanted to note that they used SDRAM, DDR and DDR2, and not DDR3 because it was not available at that time. This prompted TrueCrypt to release the following statement on their website.

Quote

“ Unencrypted Data in RAM

It is important to note that TrueCrypt is disk encryption software, which encrypts only disks, not RAM (memory).

Keep in mind that most programs do not clear the memory area (buffers) in which they store unencrypted (portions of) files they load from a TrueCrypt volume. This means that after you exit such a program, unencrypted data it worked with

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may remain in memory (RAM) until the computer is turned off (and, according to some researchers, even for some time after the power is turned off*). Also note that if you open a file stored on a TrueCrypt volume, for example, in a text editor and then force dismount on the TrueCrypt volume, then the file will remain unencrypted in the area of memory (RAM) used by (allocated to) the text editor. This applies to forced auto-dismount too.

Inherently, unencrypted master keys have to be stored in RAM too. When a non-system TrueCrypt volume is dismounted, TrueCrypt erases its master keys (stored in RAM). When the computer is cleanly restarted (or cleanly shut down), all non-system TrueCrypt volumes are automatically dismounted and, thus, all master keys stored in RAM are erased by the TrueCrypt driver (except master keys for system partitions/drives — see below). However, when power supply is abruptly

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interrupted, when the computer is reset (not cleanly restarted), or when the system crashes, TrueCrypt naturally stops running and therefore cannot erase any keys or any other sensitive data. Furthermore, as Microsoft does not provide any appropriate API for handling hibernation and shutdown, master keys used for system encryption cannot be reliably (and are not) erased from RAM when the computer hibernates, is shut down or restarted.**

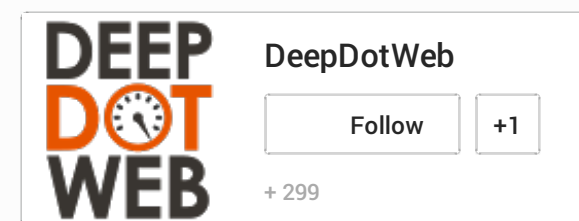
To summarize, TrueCrypt cannot and does not ensure that RAM contains no sensitive data (e.g. passwords, master keys, or decrypted data). Therefore, after each session in which you work with a TrueCrypt volume or in which an encrypted operating system is running, you must shut down (or, if the hibernation file is encrypted, hibernate) the computer and then leave it powered off for at least several minutes (the longer, the better) before turning it on again. This is required to clear

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the RAM.

* Allegedly, for 1.5-35 seconds under normal operating temperatures (26-44 °C) and up to several hours when the memory modules are cooled (when the computer is running) to very low temperatures (e.g. -50 °C). New types of memory modules allegedly exhibit a much shorter decay time (e.g. 1.5-2.5 seconds) than older types (as of 2008).

** Before a key can be erased from RAM, the corresponding TrueCrypt volume must be dismounted. For non-system volumes, this does not cause any problems. However, as Microsoft currently does not provide any appropriate API for handling the final phase of the system shutdown process, paging files located on encrypted system volumes that are dismounted during the system shutdown process may still contain valid swapped-out memory pages (including portions of Windows system files). This could cause 'blue

screen' errors. Therefore, to prevent 'blue screen' errors, TrueCrypt does not dismount encrypted system volumes and consequently cannot clear the master keys of the system volumes when the system is shut down or restarted.

<http://www.truecrypt.org/docs/unencrypted-data-in-ram>

A few key points to extract from here are that properly shutting down your computer reduces, if not completely eliminates this risk except in the case of encrypted system disks. What is meant by this is, for example, if your main operating system is Windows and you have encrypted that drive, this is your system drive and the master key for that drive is not cleared upon shutdown or restart. The solution is simply to never store anything sensitive on your system volume. Whether you use a partitioned drive or a USB stick that is encrypted, just make sure that your main drive that is booted into does not contain sensitive data. And if you have no other choice, then you need to separately encrypt the data inside the system volume with a different passphrase and private key so that even if they get into your system volume, they cannot access the other encrypted data you want to protect.

They can use these same techniques to sniff around for your PGP private key files in the RAM, so this is a very real threat in the case that if your computer is still powered on if they come to get you, they can use these techniques to

retrieve data from your computer. However, there is a debate about whether or not this type of attack can persist even now into 2014 with newer types of RAM. I point to a random blog online and I make no judgement as to whether or not this is a legitimate claim, but it is interesting nonetheless.

Quote

“ Now to test the actual cold-boot attack. Fill memory with around 1000 taint markers, just to be sure there are enough.

Now shut down. Ostensibly, the markers could be recognizable in RAM after whole minutes, but I'm impatient, so I just waited 10 seconds for the first test. Boot up, into the minimal linux installation. Load the kernel module: `insmod ./rmem.ko`. Run hunter.

Nothing.

That's ok, though. There should be at least some data corruption. The default marker size is 128 bytes, so let's set the hamming distance to 128, meaning that one bit out of every byte is allowed

to be flipped. (Statistically, that's equivalent to a 25% corruption rate, since a corrupted bit has a 50% chance of remaining the same).

Nothing.

Looks like in 10 seconds, memory was completely corrupted. Let's try a shorter interval: 2 seconds. Same results. Nothing is left of our "encryption key".

<http://bytbbox.net/blog/2013/01/cold-boot-attacks-overrated.html>

The user claimed to be using a newer type of RAM called **DDR3**, which is known to hold memory for a much shorter time than DDR2. And a newer research paper released in September 2013 tried to reproduce the findings of the 2008 research but using computers with DDR1, DDR2 and DDR3 and their findings were interesting.

Quote

“ Even though a target machine uses full disk encryption, cold boot attacks can retrieve

unencrypted data from RAM. Cold boot attacks are based on the remanence effect of RAM which says that memory contents do not disappear immediately after power is cut, but that they fade gradually over time. This effect can be exploited by rebooting a running machine, or by transplanting its RAM chips into an analysis machine that reads out what is left in memory. In theory, this kind of attack is known since the 1990s. However, only in 2008, Halderman et al. have shown that cold boot attacks can be well deployed in practical scenarios. In the work in hand, we investigate the practicability of cold boot attacks. **We verify the claims by Halderman et al. independently in a systematic fashion. For DDR1 and DDR2, we provide results from our experimental measurements that in large part agree with the original results. However, we also point out that we could not reproduce cold boot attacks against modern DDR3 chips. Our test set comprises 17 systems and system**

configurations, from which 5 are based on DDR3.

https://ieeexplore.ieee.org/xpl/login.jsp?tp=&arnumber=6657268&url=http%3A%2F%2Fieeexplore.ieee.org%2Fxppls%2Fabs_all.jsp%3Farnumber%3D6657268

So what does should you do? Number one, always shut down your computer when you are not around it or put it into hibernation mode, otherwise your sensitive documents could be lingering around in your RAM. Simply locking the screen will do you no good. Make sure your computer is using a DDR3 type of RAM, if possible. Some of you this means you need to upgrade. If you are unsure what kind of RAM your computer has, search online to find a tool that will detect it for you. Never store anything sensitive on an encrypted **system** volume, because this attack can be used to break into the volume and anything unencrypted can be retrieved. If you are using a laptop, pull the battery out so that if you need to quickly pull the power, it will turn it off immediately. If you have time, shut down the computer, otherwise turn it off immediately so that it is not running. The more time you can waste are precious seconds where they cannot retrieve any data. So immediately shut things off if you do not have enough time to do a proper shutdown.

Consider putting a lock on your computer case, and if you want to go take it a step further, bolt it to the floor. That way the amount of time it would take

them to get inside your computer would waste valuable minutes and more than likely render any recoverable memory useless. Some people have even suggested that you solder the RAM into the motherboard so they cannot take it out. This may help slow things down, but remember that cooling the memory down can preserve things for quite a while if you are using DDR1 or DDR2. With DDR3, you should be good to go and I believe with this realization, manufacturers will likely start looking at ways to encrypt RAM, but until that time you do need to be aware of this as a possible means for stealing your sensitive data and something you should keep in the back of your mind and prepare yourself for just in case.

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